DISCOUNT PROFITABILITY AUDIT

$13.8M in discounts over ten years. Did they pay for themselves?

An audit of 223,974 order lines (93,470 orders, 2016 to 2025) from the Contoso Retail dataset. This is a sample audit on synthetic data: the method carries over to a real business, the numbers do not.

$13.8M
discounts given away
6.3% of revenue and 11.3% of profit. About $101 for every discounted order line.

61%
of order lines sold below list price
Between 60.9% and 61.9% in every year from 2016 to 2025.

1 in 6
discounted sales the discount must win
If cutting discounts would lose fewer than that, the discounts cost more than they earn.

The short version

1. **Discounting is the normal price, not a promotion.**
61% of order lines are sold below list price. The rate is flat across ten years and looks the same in every product category and every sales location.

2. **It costs real money.**
$13.8M was given away: about $101 per discounted line, 6.3% of revenue, 11.3% of profit. Margin on discounted sales is 54% against 58% at full price, which is exactly what a lower price does and nothing more.

3. **Whether it pays off hinges on one number this data does not contain.**
The discounts earn their keep only if removing them would lose more than about 1 in 6 discounted sales. Nothing here shows whether this business sits above or below that line. A controlled test can.

4. **Nothing I checked explains which sales get a discount.**
Category, store, customer and year do not predict it. In a real business that is the first thing to chase down.

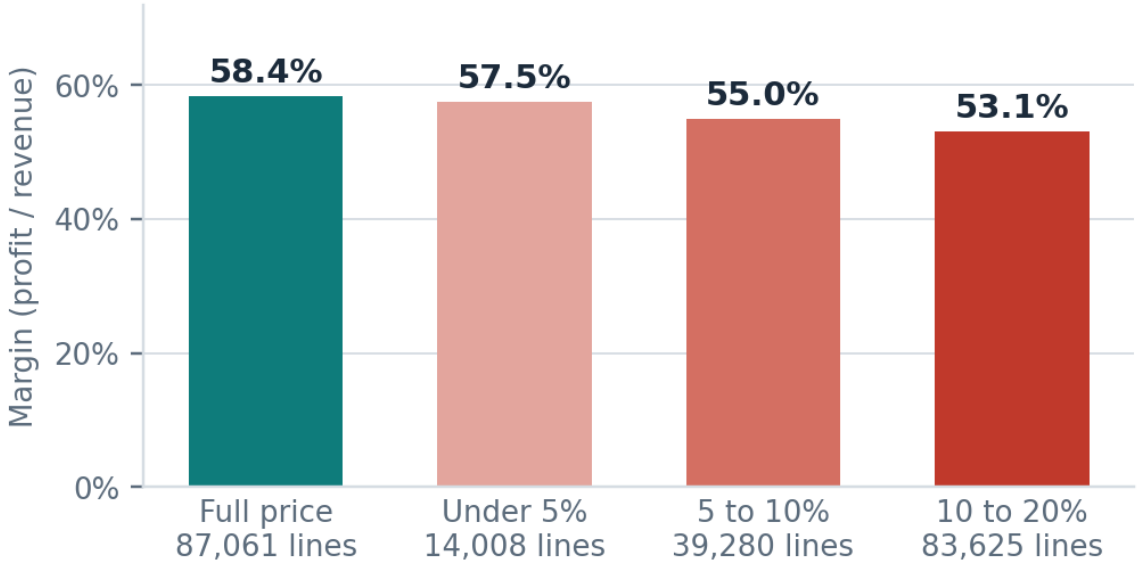
Recommended next step

Do not change pricing policy yet. Put a budget and a reason code on discounts, then run a controlled test of a lower discount cap on the 10 to 20% tier, the group that holds 61% of discounted lines. The three actions are on page 4.

Dataset at a glance. Revenue $218.8M, profit $122.3M, blended margin 55.9%. 136,913 discounted lines (61.1%) and 87,061 full-price lines. 52,189 customers. 64 sales locations: 63 stores plus the online channel, which accounts for 42% of order lines.

What the discounts do to margin

Margin falls as discounts get deeper. The fall is the price cut itself: discounted items do not cost more to buy, they sell for less.



Discount depth	Order lines	Share of discounted lines	Avg discount	Actual margin	Margin from price cut alone
Full price (no discount)	87,061	-	0%	58.4%	-
Under 5%	14,008	10%	2.5%	57.5%	57.4%
5 to under 10%	39,280	29%	7.7%	55.0%	55.0%
10 to under 20%	83,625	61%	11.8%	53.1%	52.8%
All discounted lines	**136,913**	**100%**	**9.7%**	**about 54%**	**-**

Margin = profit / revenue. 'Price cut alone' takes the 58.4% full-price margin, keeps cost unchanged and applies the tier's average discount. Actual and predicted margins agree within 0.3 points, so there is no hidden extra cost in the discounted sales. No line in the data is discounted by 20% or more.

The break-even test

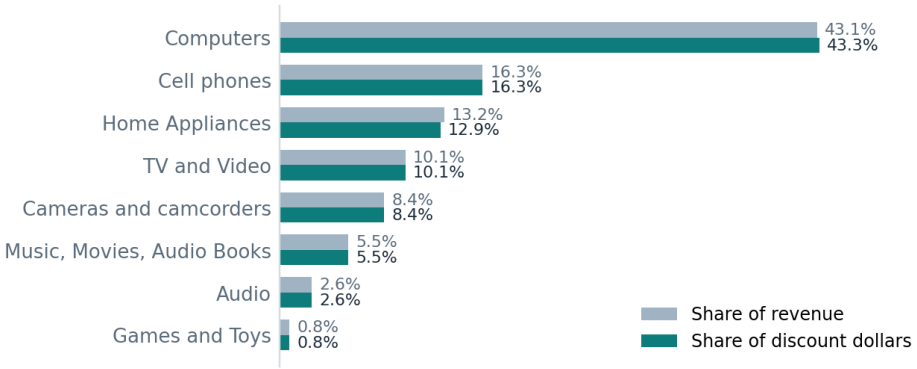
Because the data has no sales history from before the discounts, it cannot show whether discounts bring in extra sales. It can show how many would be needed.

Profit earned on discounted sales today	$69.7M
Discounts given on those sales	$13.8M
Profit on the same sales if no discount were given and every sale stayed	$83.4M
Share of those sales that must stay for removing discounts to break even	83.5%
So the discounts pay for themselves only if removing them loses more than	**16.5% (about 1 in 6)**

The calculation assumes a lost sale loses all of its profit. It ignores repeat purchases, returns and overhead. It is arithmetic, not a finding about demand: it tells you what a test has to show.

Where the discounts sit

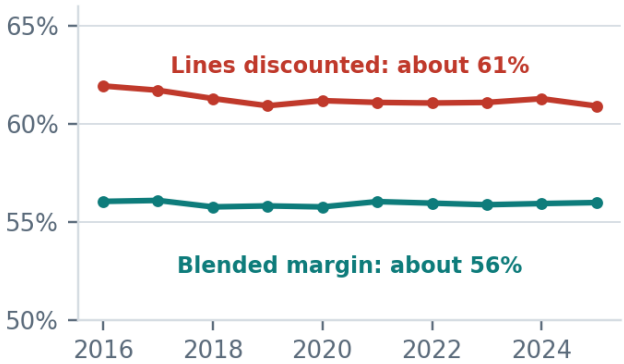
Every category is discounted the same way. The big categories carry big discount totals only because they carry big revenue.

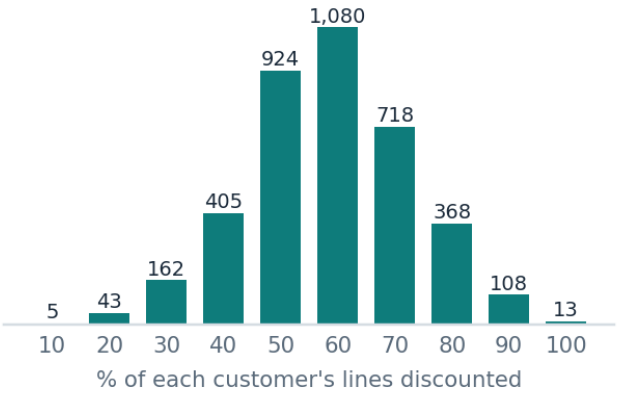


Category	Order lines	Lines discounted	Discount $	Discount as % of revenue	Share of all discount $
Computers	52,427	61.2%	$5.97M	6.3%	43.3%
Cell phones	48,287	61.0%	$2.25M	6.3%	16.3%
Home Appliances	18,674	60.9%	$1.77M	6.2%	12.9%
TV and Video	14,846	61.0%	$1.40M	6.3%	10.1%
Cameras and camcorders	14,091	61.5%	$1.16M	6.3%	8.4%
Music, Movies, Audio Books	36,645	61.4%	$0.76M	6.2%	5.5%
Audio	16,103	60.6%	$0.36M	6.3%	2.6%
Games and Toys	22,901	61.2%	$0.12M	6.4%	0.8%
All categories	**223,974**	**61.1%**	**$13.79M**	**6.3%**	**100%**

Computers and Cell phones are 45% of order lines and 59% of revenue, and they hold 60% of the discount dollars. Each category has about 61% of its lines discounted and gives up about 6.3% of its revenue to discounts.

Over time, and across customers





Same every year. The share of lines discounted stays between 60.9% and 61.9% and the blended margin between 55.8% and 56.1%, from 2016 to 2025. Nothing is getting better or worse.

Heavy buyers are not a special group. Of 3,826 customers with 10 or more order lines, 71% were discounted on 50 to 80% of their lines. Only 13 were discounted every time and none under 10%.

Same in every location. Across 64 sales locations (63 stores and online) the discount rate runs from 58.3% to 64.7%. That spread is what chance alone would produce, so no location is discounting differently from the rest. Online alone is 93,550 lines (42%) at 61.0%.

What I would do

1 **Put a budget and an owner on discounts.**
Why: Discounts cost 6.3% of revenue and the rate has not moved in ten years. That looks like a default, not a decision.
How: Set a ceiling as a share of revenue. Report discount rate, average depth and margin every week next to sales.

2 **Record why every discount is given.**
Why: Category, store, customer and year do not explain which sales get a discount (product, quantity and the promotion calendar were not tested). Without a reason on record you cannot tell discounts that win a sale from discounts that give margin away.
How: Add a required reason code with 3 to 5 options at the point of sale. After one quarter, compare profit per reason.

3 **Test a lower discount cap on the 10 to 20% tier before changing policy.**
Why: This tier is 61% of discounted lines, and its margin is 53% against 58% at full price. It is where a change matters most. Stores in this data are near-identical in discount rate and margin, so randomly assigned test and control stores are a fair comparison.
How: For 6 to 8 weeks, lower the cap in a random half of stores or weeks and leave the rest alone. Compare units sold and profit. Online is 42% of lines and the easiest place to randomize. Agree the pass mark first: removing discounts entirely breaks even at 16.5% of discounted sales lost, and a partial cap breaks even at a smaller figure, so work it out before the test starts.

What I would not do on this evidence

- **Cut discounts on Computers and Cell phones only.** They are discounted exactly like every other category. They hold 60% of discount dollars because they hold 59% of revenue.
- **Build customer-level discount tiers.** Heavy buyers are discounted at about the average rate. There is no group of customers who always, or never, get a discount.

What this data cannot tell you

- **Whether discounts bring extra sales.** There is no before-and-after or demand data. This is the question the test answers.
- **What triggers a discount.** There is no promotion, campaign or reason field.
- **True profit.** Margin is measured per order line: no overhead, returns, shipping or marketing cost.
- **Anything about a real company.** The dataset is synthetic, so patterns reflect how it was generated.

Method notes

A discounted line is one where net price differs from list price. Depth = (list price - net price) / list price. Margin = sum of profit / sum of revenue, with revenue = quantity x net price and profit = quantity x (net price - unit cost), at order-line level. Discount dollars = sum of quantity x (list price - net price). Prices are one common currency across all orders (checked against the order currency codes) and shown in USD. Tiers: under 5%, 5 to under 10%, 10 to under 20%. Closed stores were not filtered out. Built with PostgreSQL, Python and Power BI.